HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and built **Kubernetes controllers** and **CRDs** to declaratively manage stateful MongoDB deployments, integrating with the shared platform layer to provision resources.
- Developed **platform APIs** and CLIs for internal developer teams to self-provision and manage MongoDB clusters on Kubernetes, significantly improving developer experience and reducing manual intervention.
- Contributed to the design and implementation of a **developer portal** that provided discoverability and access to platform capabilities, abstracting underlying Kubernetes complexity.
- Mentored junior engineers through **code reviews** and collaborative design sessions, focusing on building predictable and maintainable platform components.
- Collaborated with Product Management and other engineering teams to translate **MongoDB features** like Search and Vector Search into Kubernetes-native APIs.
- Partnered with the **Support team** to diagnose and resolve complex customer issues related to MongoDB deployments on Kubernetes, improving resolution times by an average of **15%**.
- Containerized **MongoDB operator services** using Docker and deployed them to a Kubernetes cluster (EKS), managing configuration via Helm charts and monitoring resource utilization.

SKILLS

Kubernetes – Controllers, Operators, CRDs, Admission Webhooks, API Model, Reconciliation, Workloads, EKS

Platform Engineering – Internal Developer Platforms, Self-Service Provisioning, Platform APIs, CLIs, Developer Portals

Programming Languages – Go, Python, Java, SQL

Databases – MongoDB, PostgreSQL, MySQL, Oracle

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch)

Developer Tools – Docker, Git, Jenkins, Jira, Confluence

Distributed Systems – Stateful Workloads, Databases on Kubernetes, Production Operations

PROJECTS

Kubernetes Operator for Stateful Database Management

- Developed a Go-based Kubernetes operator to automate the deployment, scaling, and lifecycle management of stateful databases on Kubernetes, abstracting complex operational tasks.
- Implemented reconciliation loops using controller-runtime to ensure the desired state defined in custom resources (CRDs) was continuously maintained, handling over **500** database instance updates per week.
- Integrated with **AWS services** for underlying storage and networking, ensuring robust and reliable database deployments in a cloud-native environment.

Internal Developer Platform CLI Tool

- Created a Python CLI tool that interfaced with internal platform APIs, enabling developers to provision and configure development environments on Kubernetes with simple commands.
- Designed the CLI to provide self-service capabilities, reducing the average time for developers to spin up a new database instance from **4 hours** to **15 minutes**.
- Included comprehensive error handling and user feedback mechanisms to guide developers through the provisioning process and troubleshoot common issues.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)